

PROJECT DESIGN PHASE-II

Streamlining IT Procurement: Automating Standard Laptop Orders with Flow

Functional Requirements • User Stories • Data Flow • Architecture • Technology Stack

Project Area	IT Procurement Automation
Primary Workflow	Standard Laptop Request → Approval → Fulfilment
Proposed Automation	Microsoft Power Automate / Flow
Version	1.0

Purpose

This document defines a practical, auditable and scalable workflow for automating standard laptop procurement. It converts a manual request-and-email process into a structured digital flow with validation, approval routing, procurement tracking, notifications and exception handling.

1. Executive Summary

The proposed solution streamlines the ordering of standard laptops for employees. A requester submits a structured form, the automation validates the request and checks required information, routes the request to the appropriate approver, records the transaction, and sends the approved order to the IT/procurement queue. Status notifications keep the requester and stakeholders informed from submission through delivery.

- Reduce manual email-based coordination and repetitive data entry.
- Apply consistent approval rules for standard laptop requests.
- Maintain a central request register for tracking and audit.
- Provide automated confirmation, approval and fulfilment notifications.

- Create an exception path for incomplete, non-standard or rejected requests.
- Support future integration with asset, inventory, finance or procurement systems.

1.1 Scope

In Scope	Out of Scope / Future Extension
Standard laptop request submission	Complex custom hardware configuration
Requester and manager details	Full supplier contract management
Approval routing	Automated physical device deployment
Request status tracking	Warranty claims and repairs
Procurement/IT hand-off	Advanced demand forecasting
Email/Teams notifications	Direct supplier API ordering, where unavailable

Key assumption

The reference material does not specify an existing enterprise platform. The design therefore uses Microsoft Power Automate (Flow) with Microsoft Forms/SharePoint or Dataverse, Outlook/Teams and an IT/procurement queue as the proposed implementation stack. These components can be replaced with equivalent services.

2. Problem Statement & Objectives

Standard laptop procurement is often handled through email, spreadsheets and manual approvals. This creates delays, inconsistent information, duplicate requests and limited visibility into where an order is in the process. The goal is to standardize the intake and approval process while keeping human approval where it adds control.

2.1 Objectives

1. Capture complete laptop requests through a standard digital form.
2. Validate requester, department, manager, delivery location and laptop requirement before routing.
3. Automatically determine the approval path based on business rules.
4. Create a unique request ID and central record for every submission.
5. Notify the requester of submission, approval, rejection and fulfilment milestones.
6. Provide an exception path for non-standard requests or missing information.
7. Maintain an audit trail of decisions, timestamps and status changes.

3. User Journey / Experience Map

The journey below follows the employee from awareness of the need for a laptop through post-order completion. It is structured using the same experience-map concepts shown in the reference material: steps, interactions, goals, positive moments, negative moments and opportunities.

3.1 Journey stages

Stage	User action / experience	Key interaction	Goal / motivation	Potential friction
Awareness	Employee needs a standard laptop	IT portal, intranet or request link	Get the required device quickly	Not knowing where to submit
Enter	Employee opens the standard laptop request form	Microsoft Form / portal	Submit complete information once	Too many fields or unclear options
Engage	Flow validates and routes the request	Power Automate + approval	Know that the request is moving	Waiting without status visibility
Approve	Manager reviews the request	Approval card / email / Teams	Confirm business need	Approval delay or wrong approver
Fulfil	IT/procurement processes the approved request	Queue, email, procurement system	Receive device as expected	Stock/supplier delay
Exit	Employee receives device and request closes	Notification + asset record	Know when the order is complete	No delivery confirmation
Extend	Employee can review status/history	Self-service status page	Reuse a simple process next time	No history or support route

3.2 Experience flow diagram

The automation separates employee-facing actions from back-office processing. The flow can continue automatically until an approval, exception or external fulfilment dependency requires human action.

4. User Stories

User Type	Epic / Requirement	ID	User Story	Acceptance Criteria	Priority	Release
-----------	--------------------	----	------------	---------------------	----------	---------

Employee	Laptop Request	USN-01	As an employee, I want to submit a standard laptop request so that IT receives all required information.	Request is created only when mandatory fields are complete; a unique request ID is generated.	High	Sprint 1
Employee	Confirmation	USN-02	As an employee, I want confirmation after submission so that I know the request was received.	Confirmation includes request ID, summary and current status.	High	Sprint 1
Employee	Status	USN-03	As an employee, I want to see the status of my request so that I know what is happening.	Status shows Submitted, Pending Approval, Approved, In Procurement, Delivered or Rejected.	High	Sprint 1
Manager	Approval	USN-04	As a manager, I want to approve or reject a laptop request so that purchases follow business authorization.	Approval action records approver, timestamp and decision; rejection requires a reason.	High	Sprint 1
IT / Procurement	Queue	USN-05	As an IT/procurement user, I want approved requests in a queue so that I can fulfil them efficiently.	Approved requests contain requester, device standard, delivery details and approval evidence.	High	Sprint 1
IT / Procurement	Update	USN-06	As an IT/procurement user, I want to update fulfilment status so that the requester receives timely information.	Status update triggers an appropriate notification and is logged.	High	Sprint 2
Employee	Delivery	USN-07	As an employee, I want a delivery/collection notification so that I know when the laptop is ready.	Notification contains expected date or collection instructions when available.	Medium	Sprint 2
Administrator	Reporting	USN-08	As an administrator, I want a report of requests and cycle times so that the process can be monitored.	Report supports filtering by date, status, department and approver.	Medium	Sprint 3

Administrator	Exception	USN-09	As an administrator, I want non-standard requests routed for manual review so that policy exceptions are controlled.	Non-standard criteria stop automatic fulfilment and create an exception task.	Medium	Sprint 3
---------------	-----------	--------	--	---	--------	----------

5. Functional Requirements

FR No.	Functional Requirement	Sub-requirement / Rule
FR-01	Request submission	Provide a standard laptop request form with employee, department, manager, location, laptop standard and business justification fields.
FR-02	Input validation	Require mandatory fields and validate employee/manager information before submission.
FR-03	Request ID	Generate a unique request ID and store the initial status as Submitted.
FR-04	Approval routing	Route the request to the configured manager or delegated approver based on organizational data.
FR-05	Approval decision	Support Approve and Reject. A rejection must capture a reason.
FR-06	Procurement hand-off	On approval, create/update the IT/procurement queue with all relevant request details.
FR-07	Notifications	Send automated confirmations and status updates through email and/or Teams.
FR-08	Status management	Support Submitted, Pending Approval, Approved, In Procurement, Delivered, Rejected and Cancelled.
FR-09	Audit trail	Record submission, approval, rejection, status changes, timestamps and users.
FR-10	Exception handling	Route incomplete, non-standard or technically failed transactions to a manual review/exception queue.
FR-11	Closure	Close the request after delivery/collection confirmation and optionally update the asset register.

FR-12	Reporting	Provide request counts, ageing, approval time, fulfilment time and exception counts.
-------	-----------	--

5.1 Business rules

- Only authenticated employees may submit a request.
- Only standard laptop models defined by IT are eligible for the automated path.
- Every request must have an accountable approver.
- A rejected request cannot proceed to procurement without a new submission or authorized override.
- Non-standard hardware, missing inventory or failed integrations must enter the exception path.
- A request should not be considered complete until fulfilment/delivery is recorded.

6. Non-Functional Requirements

NFR No.	Requirement	Description / Target
NFR-01	Usability	The form should be simple, mobile-friendly where applicable, and use clear labels, defaults and validation messages.
NFR-02	Security	Use organizational authentication, least-privilege access, protected connections and restricted access to procurement records.
NFR-03	Reliability	Failures should be logged and retried where safe; unresolved failures should create an exception alert.
NFR-04	Performance	Normal form submission should acknowledge quickly; automated routing should occur without unnecessary manual delay.
NFR-05	Availability	The workflow should rely on supported cloud services and avoid single-user/manual dependencies.
NFR-06	Scalability	The design should support increased request volume by separating intake, automation, data and fulfilment integrations.

NFR-07	Auditability	All important decisions and status changes should have timestamps, request IDs and actor information.
NFR-08	Maintainability	Approval thresholds, standard laptop catalogue and notification templates should be configurable rather than hard-coded.

7. Data Flow Diagram

The data flow shows how request information enters the solution, is validated and routed, stored for audit, and passed to IT/procurement. The employee and approver provide human inputs; the automation coordinates the system-to-system movement of information.

7.1 Main data flows

Flow	Source	Destination	Data
Request submission	Employee	Request Flow	Employee ID, department, manager, location, laptop standard, justification
Validated request	Request Flow	Approval service	Request ID, requester details, selected standard, business justification
Approval decision	Manager	Request Flow	Decision, approver, timestamp, comments
Procurement hand-off	Request Flow	IT/Procurement	Approved request, delivery information, required-by date
Status update	IT/Procurement	Request Register	Procurement status, order/reference, expected delivery, completion
Notification	Request Flow	Employee / Manager	Status, request ID, action required, next step
Audit record	Request Flow	Request Register	Event type, actor, timestamp, previous/new status

8. Technical Architecture

The proposed architecture uses a low-code automation layer so that procurement rules can be changed without rewriting a complete application. A central request register provides a system of record, while Microsoft 365 services provide communication and approval experiences.

8.1 Component responsibilities

Component	Responsibility	Proposed Technology
User Interface	Collect request information and display instructions	Microsoft Forms or Power Apps
Application Logic	Validate, route, update status and handle exceptions	Microsoft Power Automate
Approval	Manager decision and comments	Power Automate Approvals / Teams
Data Store	Request register, status, audit fields	SharePoint Lists or Dataverse
Communication	Confirmation and status messages	Outlook / Microsoft Teams
Identity	Authentication and employee/manager identity	Microsoft Entra ID
Procurement Queue	Approved requests awaiting fulfilment	SharePoint/Dataverse view or procurement system
Asset Integration	Optional device/asset update	Existing IT asset management API
Monitoring	Flow run history, failures and alerts	Power Automate monitoring + admin reporting

9. Technology Stack

S.No.	Component	Description	Technology
1	User Interface	Employee request experience	Microsoft Forms / Power Apps
2	Application Logic	Business logic and orchestration	Microsoft Power Automate (Cloud Flows)
3	Approval	Manager authorization	Power Automate Approvals / Teams
4	Database / Register	Request, status and audit information	SharePoint Lists or Dataverse

5	Notifications	Email and collaboration messages	Microsoft Outlook / Teams
6	Identity & Access	Authentication and role-based access	Microsoft Entra ID
7	File Storage	Optional quotations, supporting documents	SharePoint Document Library
8	External API	Optional procurement/asset integration	REST API / approved connector
9	Analytics	Operational reporting	Power BI / SharePoint views
10	Infrastructure	Cloud execution and connectors	Microsoft Power Platform

9.1 Application characteristics

Characteristic	Design approach	Technology / Control
Open-source frameworks	Not required for the low-code implementation; custom extensions can use standard web technologies where necessary.	Power Platform; optional HTML/CSS/JavaScript for custom UI
Security implementation	Authenticated users, least privilege, connector permissions and controlled sharing.	Entra ID, Power Platform roles, SharePoint/Dataverse permissions
Scalable architecture	Separate intake, orchestration, data and integration responsibilities.	Cloud Flows, Dataverse/SharePoint, APIs
Availability	Use managed cloud services and monitored flows; avoid manual single points of failure.	Power Platform + monitoring
Performance	Minimize unnecessary connector calls, validate early and use asynchronous notifications.	Power Automate optimization, filtered queries, retries

10. Detailed Workflow

8. Employee opens the Standard Laptop Request form.
9. Employee selects the standard laptop category and provides required business/delivery details.
10. The flow validates mandatory fields and basic eligibility.

11. The flow creates a request record and generates a unique Request ID.
12. The employee receives a submission confirmation.
13. The flow identifies the employee's manager or configured approver.
14. The approver receives an approval request with the essential request summary.
15. If rejected, the flow records the reason, changes the status to Rejected and notifies the employee.
16. If approved, the flow records the approval evidence and sends the request to the IT/procurement queue.
17. IT/procurement processes the request and updates order/fulfilment information.
18. The employee receives status notifications at configured milestones.
19. After delivery or collection, IT records completion and the flow closes the request.
20. The final status and audit history remain available for reporting.

10.1 Exception paths

Exception	System response	Human action
Missing required data	Prevent submission or request correction	Employee corrects the request
No approver found	Create exception and notify IT/admin	Admin assigns an approver
Non-standard laptop	Stop automated fulfilment	IT reviews and manually processes
Approval timeout	Send reminder; escalate according to policy	Approver acts or escalation owner intervenes
Connector/API failure	Retry where safe; log failure	Support investigates unresolved failure
No stock	Move to procurement/backorder status	Procurement confirms supply/ETA
Duplicate request	Flag likely duplicate based on configurable rules	IT/admin confirms whether to cancel/merge

11. Security, Privacy & Audit

- Authentication: require organizational sign-in for employee and approver actions.
- Authorization: employees see their own requests; approvers see assigned approvals; IT/procurement sees operational requests; administrators have controlled elevated access.
- Data minimization: collect only information required to fulfil and audit the request.
- Sensitive information: avoid storing passwords, unnecessary personal information or confidential procurement data in notification messages.

- Audit: retain request ID, actor, timestamp, status transition and decision reason where required.
- Connector governance: use approved service connections and avoid personal accounts as the owner of production flows.
- Error handling: failure notifications should not expose confidential request data unnecessarily.
- Retention: apply the organization's records-retention and deletion requirements to request and audit data.

Production control

Before deployment, the organization should confirm its Power Platform environment strategy, data residency/retention rules, connector licensing, DLP policies, approval delegation rules and ownership model.

12. Testing & Acceptance

Test ID	Scenario	Expected Result	Priority
TC-01	Submit complete standard laptop request	Request ID generated and confirmation sent.	High
TC-02	Submit without a mandatory field	Submission blocked with a clear validation message.	High
TC-03	Manager approves request	Status changes to Approved and procurement item is created/updated.	High
TC-04	Manager rejects request	Status becomes Rejected; reason is stored and employee is notified.	High
TC-05	No manager/approver available	Exception is created and routed to administrator.	High
TC-06	Non-standard device selected	Automatic path stops and manual review is triggered.	High
TC-07	Procurement changes status	Request register updates and appropriate notification is sent.	High
TC-08	Delivery confirmed	Request closes and completion is recorded.	High

TC-09	Automation connector fails	Retry/exception behavior occurs and support is alerted.	Medium
TC-10	Unauthorized user attempts access	Access is denied according to role permissions.	High
TC-11	Duplicate request detected	Request is flagged for review instead of creating uncontrolled duplicate work.	Medium
TC-12	Reporting query	Counts and cycle times match request-register records.	Medium

13. Success Metrics / KPIs

KPI	Definition	Target Direction
Request completion time	Time from submission to procurement hand-off	Decrease
Approval turnaround time	Time from approval request to decision	Decrease
First-time completeness	Percentage of requests submitted without correction	Increase
Automation rate	Percentage of standard requests completed without manual exception	Increase
Exception rate	Percentage of requests routed to manual review	Decrease, where appropriate
Status visibility	Percentage of active requests with a current status	Increase
Notification success	Percentage of configured notifications delivered successfully	Increase
Audit completeness	Percentage of requests with required decision/status evidence	Increase

14. Implementation Roadmap

Release	Scope	Deliverables
---------	-------	--------------

Sprint 1	Core intake and approval	Request form, validation, request register, Request ID, manager approval, basic notifications
Sprint 2	Procurement fulfilment	IT/procurement queue, status updates, delivery notification, exception handling
Sprint 3	Reporting and integration	Dashboard, cycle-time metrics, asset/procurement integration, duplicate detection
Future	Optimization	Self-service status page, inventory availability check, supplier integration, advanced analytics

14.1 Roles & responsibilities

Role	Responsibilities
Employee / Requester	Submit accurate request, respond to corrections, confirm delivery/collection.
Manager / Approver	Review business need and approve/reject within policy.
IT / Procurement	Validate fulfilment, place/track order, update status and close request.
Flow Administrator	Maintain automation, connections, rules, error handling and monitoring.
System Administrator	Manage platform environments, access, governance and security controls.
Business Owner	Define procurement policy, approval rules, standards and success measures.

15. Conclusion

Automating standard laptop orders with Flow provides a structured path from employee request to approval and fulfilment. The proposed design keeps the process simple for employees while introducing validation, consistent approvals, centralized tracking, automated notifications and an auditable exception path. The architecture is intentionally modular so that a SharePoint/Dataverse register can later be integrated with inventory, asset management, procurement or finance systems without redesigning the employee journey.

Recommended next step

Validate the proposed fields, approval rules, standard laptop catalogue, procurement hand-off method and retention requirements with the IT procurement process owner. Once confirmed, build the Sprint 1 minimum viable flow and test it with a small pilot group.

Appendix A – Suggested Request Form Fields

Field	Type	Required	Example / Rule
Employee name / ID	User lookup	Yes	Authenticated employee
Department	Choice / lookup	Yes	Pre-populated where possible
Manager	User lookup	Yes	Derived from directory where possible
Work location	Choice	Yes	Office / Remote / Site
Laptop standard	Choice	Yes	Standard / Standard Plus / Developer, if approved
Business justification	Long text	Yes	Reason for new/replacement device
Required-by date	Date	Yes	Policy-defined date range
Delivery / collection preference	Choice	Yes	Delivery / Office collection
Additional notes	Long text	No	Optional operational notes
Supporting document	Attachment	No	Only when policy requires it

Appendix B – Status Model

Status	Meaning	Owner / Next Action
Submitted	Request successfully created	Automation validates
Pending Approval	Awaiting approver decision	Manager / delegate
Approved	Approved for fulfilment	IT / Procurement
In Procurement	Order/fulfilment is underway	IT / Procurement
Backordered	Approved but waiting for stock	Procurement

Delivered	Device delivered/collected	IT closes request
Rejected	Approval denied	Employee may resubmit if needed
Cancelled	Request withdrawn or cancelled	Requester / IT
Exception	Manual investigation required	IT / Administrator